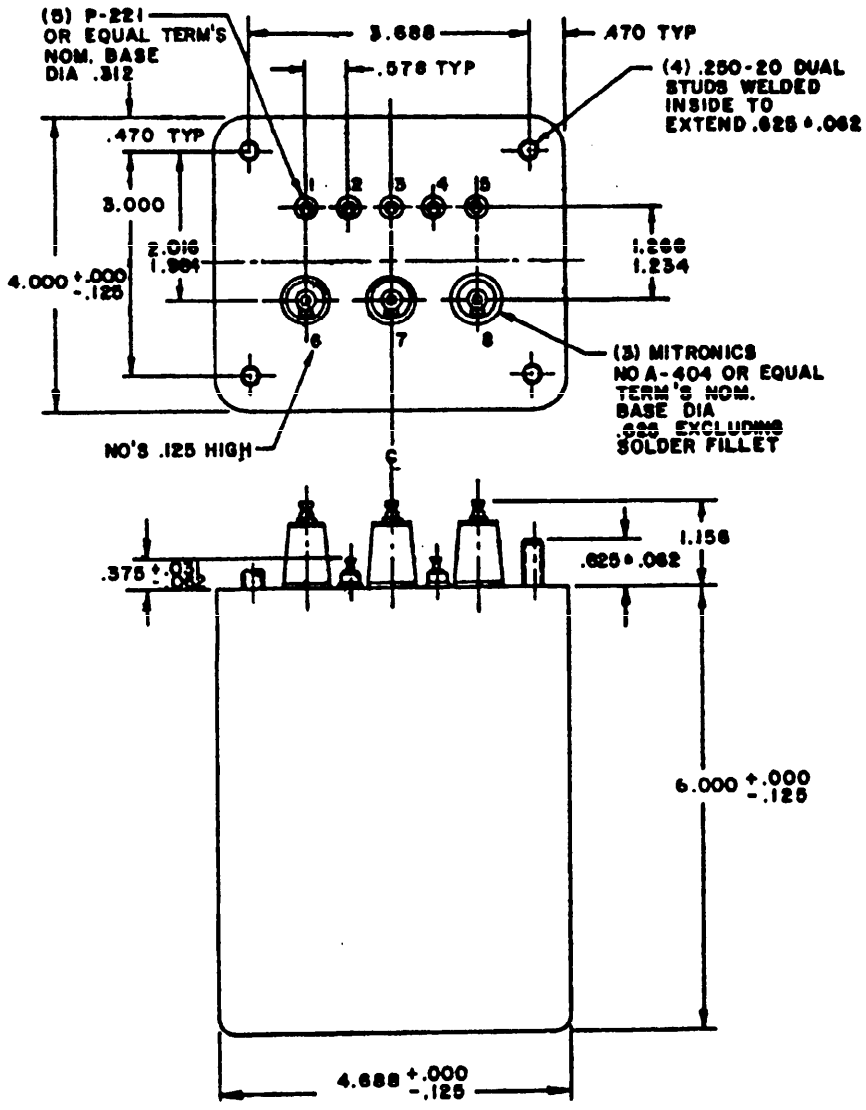
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-UP, TF4RO3MA932

This specification is approved for use by all Departments and Agencies of the Department of Defense.

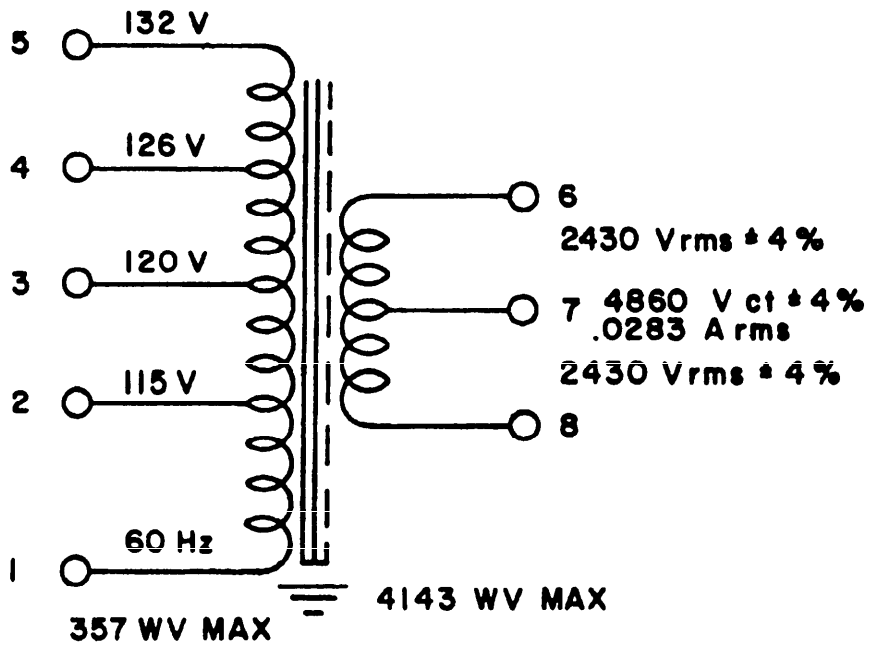
The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.



INCHES	MM
.031	.79
.062	1.57
.125	3.18
.250	6.35
.312	7.92
.375	9.53
.470	11.94
.562	14.27
.625	15.88
1.156	29.36
1.234	31.34
1.266	32.16
1.984	50.39
2.016	51.21
3.000	76.20
3.688	93.68
4.000	101.60
4.688	119.08
6.000	152.40

FIGURE 1. Dimensions and configuration.

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CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.030 (.76 mm) for three place decimals.
4. Marking shall be on the end opposite terminals.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding).

Electrical ratings:
Volt ampere: 160 VA.
Primary voltage,
(1-2): 115 volts, 60 Hz, ±10 percent.
(1-3): 120 volts, 60 Hz, ±10 percent.
(1-4): 126 volts, 60 Hz, ±10 percent.
(1-5): 132 volts, 60 Hz, ±10 percent.
Secondary voltage:
(6-8): 4860 volts ct, ±4 percent, .0283 ampere rms.
DC resistance:
(1-5) 1.5 ohms nominal.
(6-8) 2400 ohms nominal.
Working voltage:
(1-5): 357 volts maximum.
(6-8): 4143 volts maximum.

Design and construction:

Dimensions and configuration: See figure 1.
Duty cycle: Continuous.
Case: Fully enclosed, hermetically sealed.
Material: Steel.
Terminals: Standoff, solder type.
Terminal height:
(1 thru 5): .375 ±.031.
(6 thru 8): 1.156 ±.031.
Weight: 15 pounds nominal.
Operating temperature range: -55°C to 105°C.
Altitude: 15,000 feet maximum.

Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds.

Dielectric withstanding voltage:
Primary: 1,000 volts rms.
Secondary: 6,800 volts rms.

At barometric pressure:
Primary: 1000 volts rms.
Secondary: 5200 volts rms.

Electrical characteristics:
No load: With 115 volts, 60 Hz across (1-2), current in (1-2), shall not exceed 550 mA and the power in (1-2) shall not exceed 16 watts.
Voltage across (6-8): 5,020 volts ct, +2 percent.
Rated load: With 115 volts, 60 Hz across (1-2), the voltage across (6-8) shall be 4860 volts ct, ±4 percent.
Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1.
Magnetic shielding: Applicable.
Temperature rise: 40°C with 132 volts rms, 60 Hz across (1-5) at an ambient temperature of 65°C.

Marking location: See figure 1.
Part number: M27/113-01.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, SC
Navy - EC, SM, CS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - MC, WC, SC, ME
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5950-0520)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-27/113 TRANSFORMER, POWER, STEP-UP, TF4RO3MA932

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-PM 1068-68

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U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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